

Euclidean geometry – 2

Conics, areas and projective maps

Homework

24 сентября 2026 года

Theoretical minimum. The conic with focus F , directrix l and eccentricity e is $\{X : |XF| = e d(X, l)\}$ (ellipse, parabola, hyperbola for $e < 1$, $e = 1$, $e > 1$); in polar coordinates at F it is $r = p/(1 + e \cos \varphi)$. For $e \neq 1$: $|XF_1| \pm |XF_2| = \text{const} = 2a$, $c = ea$, directrices at distance a/e from the centre. The tangent bisects the angle between the focal radii (externally for the ellipse), so rays from one focus are reflected into the other. Normals to $x^2 = 2py$ through (u, v) have feet t with $t^3 + 2p(p-v)t - 2p^2u = 0$. Up to affine maps a non-degenerate conic is $x^2 + y^2 = 1$, $x^2 - y^2 = 1$ or $y = x^2$; five points, no four collinear, lie on a unique conic, and the conics through four points form a pencil $\lambda Q_1 + \mu Q_2$ with three line pairs. $[ABC] = \frac{1}{2} \det(B - A, C - A)$ is affine in each vertex; affine maps multiply areas by $\det M$. Perspectivities preserve the cross-ratio $(a, b; c, d) = \frac{(c-a)(d-b)}{(c-b)(d-a)}$; projective maps $x \mapsto \frac{\alpha x + \beta}{\gamma x + \delta}$ are fixed by three points and are compositions of perspectivities. The inscribed n -gon of maximal perimeter in a smooth strictly convex curve is a billiard trajectory.

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1. Let A be the area of the region in the first quadrant bounded by the line $y = \frac{1}{2}x$, the x -axis and the ellipse $\frac{1}{9}x^2 + y^2 = 1$. Find the positive number m such that A is equal to the area of the region in the first quadrant bounded by the line $y = mx$, the y -axis and the ellipse. (Putnam 1994 A2)
 2. A *standard parabola* is the graph of a quadratic polynomial $y = x^2 + ax + b$ with leading coefficient 1. Three standard parabolas with vertices V_1, V_2, V_3 intersect pairwise at points A_1, A_2, A_3 (the ones with vertices V_j, V_k at A_i , $\{i, j, k\} = \{1, 2, 3\}$). Let $A \mapsto s(A)$ be the reflection of the plane in the x -axis. Prove that the standard parabolas with vertices

$s(A_1), s(A_2), s(A_3)$ intersect pairwise at the points $s(V_1), s(V_2), s(V_3)$.
(IMC 2002)

3. Suppose that the section of the cone $x^2 + y^2 = z^2, z \geq 0$, by a plane is an ellipse. Prove that the origin is a focus of the orthogonal projection of this ellipse to the plane Oxy . (МФТИ 2015)
4. Let $a > b > 0$ and let ε be the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Let F be one of its foci, and let A and B be points of ε such that F lies on the segment AB and AB is not the major axis. Let C be the intersection point of the tangents to ε at A and B . Prove that (a) $CF \perp AB$; (b) C lies on a directrix of the ellipse. (OMOUS 2023)
5. An ellipse with foci F_1 and F_2 lies inside a circle with centre O and radius r and touches it at two points. Prove that the eccentricity of the ellipse equals OF_1/r . (Мехмат, матбой 2010)
6. Let A and B be points on the same branch of the hyperbola $xy = 1$. Suppose that P is a point lying between A and B on this hyperbola, such that the area of the triangle APB is as large as possible. Show that the region bounded by the hyperbola and the chord AP has the same area as the region bounded by the hyperbola and the chord PB . (Putnam 2015 A1)
7. Let A, B be different points on a parabola. Prove that one can find points $P_1, P_2, \dots, P_n$ between A and B on the parabola such that the area of the convex polygon $AP_1P_2 \dots P_nB$ is maximal. Prove that in this case the ratio of this area to the area of the parabolic segment cut off by AB does not depend on A and B but only on n , and find it. (IMS 2008)
8. Let a, b, c be three points of a parabola, b lying on the arc between a and c , and let A, B, C be the intersection points of the tangents at b and c , at c and a , at a and b , respectively. Denote by S_{xyz} the area of the triangle xyz . Prove that (a) the circumcircle of ABC passes through the focus of the parabola; (b) the orthocentre of ABC lies on the directrix; (c) $S_{abc} = 2S_{ABC}$; (d) $\sqrt[3]{S_{abC}} + \sqrt[3]{S_{bcA}} = \sqrt[3]{S_{caB}}$. (OMOUS 2024)
9. Let A, B, C, D be four distinct points of a non-degenerate conic. Prove

that the cross-ratio of the lines XA, XB, XC, XD (i.e. of their intersection points with a fixed line) is the same for all points $X \neq A, B, C, D$ of the conic (Chasles, Steiner). (Hint: map the conic to a circle by a projective map and use inscribed angles.) Deduce Pascal's theorem: for a hexagon inscribed in a non-degenerate conic, the three intersection points of pairs of opposite sides (possibly at infinity) lie on one line.

10. Let $A_1A_2 \dots A_{3n}$ be a closed broken line consisting of $3n$ line segments in the Euclidean plane. Suppose that no three of its vertices are collinear, and for each index $i = 1, 2, \dots, 3n$, the triangle $A_iA_{i+1}A_{i+2}$ has counterclockwise orientation and $\angle A_iA_{i+1}A_{i+2} = 60^\circ$, using the notation $A_{3n+1} = A_1$ and $A_{3n+2} = A_2$. Prove that the number of self-intersections of the broken line is at most $\frac{3}{2}n^2 - 2n + 1$. (Crossings of segments are also counted in Combinatorics – 6.) (IMC 2014)